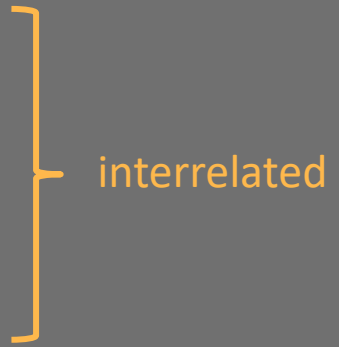
Integrating DB & Application

Some issues

- How to connect to database
- How to send SQL commands
- How to communicate data to/from DB
- What code goes where





Connecting to database

Connection string

postgresql://riceuser:ricepassword@localhost/ricedb

- Used by many libraries.
- Can often specify pieces separately.

Key pieces:

- Driver
- User name
- User password
- DB server
- DB name

Demo: Connection string

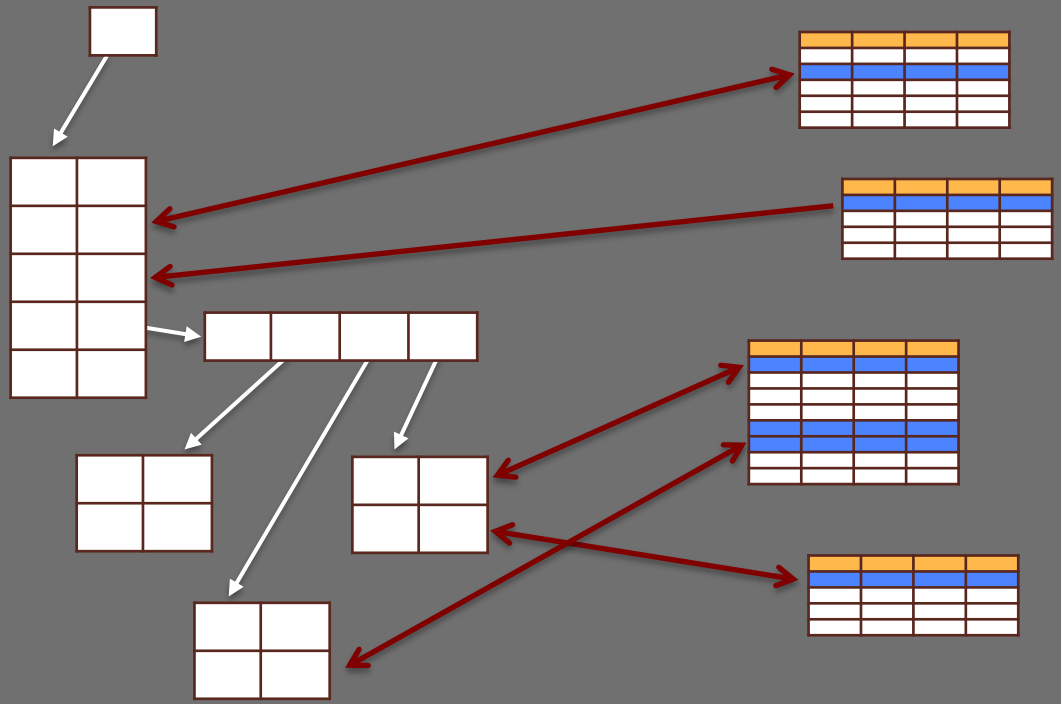
- Show connection string construction in Python & Java code examples.

Sending commands to DB

Two basic styles:

- Simple API, plain SQL
- Complex API (Object-Relational Mapping)

Mapping data is HARD



Object-Relational
Impedance Mismatch

Mapping data is HARD

Under the hood, often involves two mappings:

- Application types → string
- string → DB types

Communicating, parsing, & unparsing data is relatively expensive.

How to organize code?

